Coupling Polytopes and Toric Geometry: the 2×3 Family, the Square Case, Higher Levels and Stability

A companion to *Conditionals, Couplings and Toric Geometry*

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Abstract

A coupling of two finite distributions is a point of a transportation polytope. For nondegenerate marginals this polytope is Delzant [2], so it is the moment polytope of a compact toric symplectic manifold. This companion works the correspondence out in the cases that can be computed completely, and records how far it goes.

- *The* 2×3 *family.* The six walls $f_i = g_j$ cut the space of marginals into 18 chambers, which fall into 5 orbits under the symmetry group $S_2 \times S_3$. The surfaces that occur are $\mathbb{CP}^2$, $\mathbb{CP}^1 \times \mathbb{CP}^1$, the Hirzebruch surface F_1, $\mathbb{CP}^2$ blown up at two points, and the del Pezzo surface of degree 6. Every facet normal lies in the fan of the last, so every surface in the family is the del Pezzo surface or a toric blow-down of it. Crossing a single wall is exactly one toric blow-up or blow-down.
- *The Kähler class.* It is the product coupling read as divisor coefficients, $[\omega_{f,g}] = \sum_c f_i g_j [D_c]$. Edge lengths of the coupling polygon are degrees of ω on toric curves, and its area is the symplectic volume.
- *The square* 2×2 *case.* With the unital whiskering, explicit whiskering defects are nonzero, and every defect is a tangent vector of a coupling polytope.
- *Higher levels.* The Delzant theorem applies at levels two and three. Flattening a level-two coupling is affine and commutes with stochastic change of base, but it is neither surjective nor injective in general.
- *Disc potentials.* They are sums over facets, which are cells, not cycles. For the degree-6 del Pezzo surface the potential has six critical points, all over the product coupling, with critical values $6, -2, -2, -2, -3, -3$. Square nondegenerate polytopes with $n \geq 3$ give manifolds that are not Fano. Optimal transport minimizes a moment-map component, so its optima are torus-fixed points, that is, spanning-tree couplings.
- *Stability.* Each $\omega_{f,g}$ gives a Bridgeland stability condition on the derived category of the corresponding surface, with explicit central charges. We state what this does and does not connect.

*Code and checks: `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/coupling`. Verification script: `check_polytope_companion.py` (26 checks; exact rational arithmetic except where marked). This version replaces the draft "Coupling polytopes and the 2×3 family"; appendix A lists what was corrected and what was withdrawn.

- *Cumulants.* Centred couplings are second mixed cumulants, and level-two flattening keeps only the cross-covariance of random measures. The whiskering defects are the Boolean cumulants of whiskering, viewed as a noncommutative expectation. They are circulations, so tangent vectors of coupling polytopes. A cone in the sense of homotopy probability theory makes whiskering an A_∞-functor through arity three, and its cohomology keeps only the marginals. Into the square-zero cone, whiskering is an A_∞-functor with exactly two components.
- *The coupling nerve.* Couplings form the simplicial object $\mathsf{D}(Y^{\bullet+1})$. Its 2-horns always fill, and the fillers form products of coupling polytopes, with the gluing filler the maximum-entropy one. Its 3-horns and 4-horns can fail to fill, with contextual fraction $\frac{1}{2}$ and $\frac{3}{4}$, and $1-2^{2-n}$ in general, so it is not Kan. Boundary fibres are Delzant for shape $2\times n\times m$, orbifold polytopes for $3\times 3\times 3$, and eventually arbitrary.
- *The tower as a fibration.* In a worked example, transport along kernels keeps a failure, destroys it (constant kernel), pulls certificates back, never increases the secondary obstruction, and bounds the distances at all three levels by the distance between kernels.
- *The family over marginals.* Inside a chamber the polytopes vary linearly. Across a wall the Duistermaat–Heckman volume is C^1 with a jump in its second derivative, while the Fisher metric at the product coupling is smooth: walls are toric, not statistical.
- *The open problems.* The numbers behind the theory paper's experimental definitions are recorded here: approximate fillers of parity horns up to $n = 8$; wall counts and crossing types for 2×4 and 3×3; glued central charges across the pentagon–hexagon wall, where $\mathrm{area}_+ = \mathrm{area}_- - \frac{1}{2}s^2$; and the Floer-nontrivial fibres. The product fibre is proved Floer-trivial for generic marginals; at the four sampled points the nontrivial fibres lie over the leximin coupling and two signed-balanced couplings (a computation, not a theorem).
- *The tower and the nerve.* Encoded as tower claims, the parity horns pass the local test but cannot be lifted. Their secondary obstruction is twice the contextual fraction, and their distance to the fillable horns is twice the best approximate filler distance. The Fisher bound on the obstructions is attained in the worked tower example.
- *Synthetic data.* A classifier's confusion table is a coupling. The tables cover: chamber identification from samples against its Hoeffding bound; independence tests, whose two degrees of freedom are the dimension of the hexagon; two models scored on one label set, whose correlated errors the gluing filler misses by a factor of six; and horn data from separate studies, where sampling noise and contextuality separate.

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1 Introduction

The companion paper [2] proves that the polytope $\mathrm{Cpl}(f, g)$ of couplings of two nondegenerate marginals is Delzant. By Delzant's theorem [10] there is therefore a compact toric symplectic manifold $X_{f,g}$ whose moment map has image $\mathrm{Cpl}(f, g)$. Each interior coupling is the image of a Lagrangian torus, and each vertex is the image of a fixed point. This note asks what the correspondence gives in concrete cases.

Section 2 fixes conventions and proves a general formula for the Kähler class. Section 3 treats the 2×3 family completely. Section 4 treats the square 2×2 case, the base case for the whiskering questions of [2, §§5, 9]. Section 5 extends the polytope results to second- and third-order states. Section 6 computes disc potentials and relates optimal transport to the moment map. Section 7 constructs stability conditions from marginals and states the remaining questions. Section 8 records the status of the A_∞ question, section 9 connects the polytopes to cumulants and homotopy probability theory, section 10 computes the low-dimensional fillers of the coupling nerve, section 11 follows one fibre of the tower along explicit kernels, and section 12 describes how the polytopes vary with the marginals. Every numbered claim is checked in `check_polytope_companion.py`.

2 Setup

Let $f \in \mathsf{D}(Y)$ and $g \in \mathsf{D}(Y')$ have full support, with $|Y| = n$ and $|Y'| = m$, and let

$$\mathrm{Cpl}(f,g) = \Big\{\alpha \in \mathbb{R}^{Y\times Y'}_{\geq 0} : \ \textstyle\sum_{y'} \alpha(y,y') = f(y),\ \sum_{y} \alpha(y,y') = g(y')\Big\}.$$

The tangent space is the space of circulations, that is matrices with zero row and column sums. The integer circulations form a lattice Λ of rank $(n-1)(m-1)$ with basis $e_{ij} = E_{ij} - E_{im} - E_{nj} + E_{nm}$ $(i < n,\ j < m)$. Coordinates in this basis are the entries in the top-left $(n-1)\times(m-1)$ block. The pair (f,g) is *degenerate* if $\sum_I f = \sum_J g$ for some nonempty proper $I \subsetneq Y$ and $J \subsetneq Y'$. These hyperplanes are the *walls*, and the components of their complement are the *chambers*.

Theorem 2.1 ([2, Thm. 6.4])**.** *If (f,g) is nondegenerate, every vertex of $\mathrm{Cpl}(f,g)$ is supported on a spanning tree of $K_{n,m}$, the edges at a vertex are the fundamental cycles of that tree, and $\mathrm{Cpl}(f,g)$ is Delzant with respect to Λ.*

Facets and normals. The facets are among the sets $\{\alpha_c = 0\}$, one for each cell $c = (i,j)$. A cell gives no facet when $f_i + g_j > 1$, since then $\alpha_c \geq f_i + g_j - 1 > 0$. The inward normal of $\{\alpha_c = 0\}$ is the coordinate functional $\alpha \mapsto \alpha_c$ restricted to Λ. In the dual basis $\{e^*_{ij}\}$ it is

$$v_{(i,j)} = e^*_{ij}, \quad v_{(i,m)} = -\textstyle\sum_j e^*_{ij}, \quad v_{(n,j)} = -\sum_i e^*_{ij}, \quad v_{(n,m)} = \sum_{i,j} e^*_{ij} \qquad (i < n,\ j < m).$$

Each v_c is primitive, because some integer circulation has entry 1 at c.

Proposition 2.2 (The Kähler class is the product coupling)**.** *For nondegenerate (f,g), let D_c be the toric divisor of the facet $\{\alpha_c = 0\}$. Up to the normalization of ω, the symplectic class of $X_{f,g}$ is*

$$[\omega_{f,g}] = \sum_{c \ facet} f_i g_j \, [D_c], \qquad c = (i,j),$$

and $\omega_{f,g} \cdot C$ is the lattice length of the edge of $\mathrm{Cpl}(f,g)$ that corresponds to a torus-invariant curve C.

Proof. For a Delzant polytope $\{u : \langle u, v_c\rangle + \lambda_c \geq 0\}$ the symplectic class is $\sum_c \lambda_c [D_c]$ [14]. Here the facet functions are $\alpha_c(u) = \langle u - u_0, v_c\rangle + \alpha_c(u_0)$ for any interior u_0, so $\lambda_c = \alpha_c(u_0)$. Taking $u_0 = f \otimes g$ gives the formula. Another base point changes the coefficients by $\langle w, v_c\rangle$, which is a linear relation among the $[D_c]$. The degree of ω on a torus-invariant curve is the lattice length of its edge. □

3 The 2×3 family

Write $f = (f_1, 1-f_1)$, $g = (g_1, g_2, 1-g_1-g_2)$, $x = \alpha_{11}$ and $y = \alpha_{12}$. The polygon $\mathrm{Cpl}(f,g)$ is cut out by

$$x \geq 0, \quad y \geq 0, \quad x \leq g_1, \quad y \leq g_2, \quad x + y \leq f_1, \quad x + y \geq f_1 + g_1 + g_2 - 1,$$

with inward normals e_1, e_2, $-e_1$, $-e_2$, $-(e_1+e_2)$ and e_1+e_2. The only proper subsets of a two-point and a three-point set that can have equal sums give the six walls $f_i = g_j$.

Theorem 3.1 (Chambers and surfaces)**.** *The six walls cut the prism $\{0 < f_1 < 1,\ g_1, g_2 > 0,\ g_1 + g_2 < 1\}$ into 18 chambers. On each chamber the polygon type is constant and Delzant, and the surfaces are:*

Polygon	*Chambers*	*Self-intersections* D_c^2	*Surface*	*Orbit under* $S_2 \times S_3$
triangle	*2*	$(1,1,1)$	$\mathbb{CP}^2$	*one orbit of size 2*
quadrilateral	*3*	$(0,0,0,0)$	$\mathbb{CP}^1 \times \mathbb{CP}^1$	*one orbit of size 3*
quadrilateral	*6*	$(1,0,-1,0)$	$F_1 = \mathrm{Bl}_1\mathbb{CP}^2$	*one orbit of size 6*
pentagon	*6*	$(0,-1,-1,-1,0)$	$\mathrm{Bl}_2\mathbb{CP}^2$	*one orbit of size 6*
hexagon	*1*	$(-1)^6$	*del Pezzo of degree 6*	*fixed*

Here S_2 swaps the two rows and S_3 permutes the columns.

Proof. For an arrangement of hyperplanes restricted to a convex region, each realizable sign vector is exactly one convex chamber. A rational grid and random sampling realize the same 18 sign vectors. In each chamber the vertices are computed exactly, and the number of vertices and the self-intersections are constant on all samples. Self-intersections come from the relations $v_{i-1}+v_{i+1} = -D_i^2 v_i$ between cyclically ordered normals. Consistency is confirmed by $\sum_i D_i^2 = 12 - 3k$ for k divisors. Delzant means that consecutive normals have determinant ± 1. The orbits are computed by letting $S_2 \times S_3$ act on sign vectors. □

Theorem 3.2 (Every surface is a blow-down of the del Pezzo surface)**.** *For every nondegenerate pair in the 2×3 family, $X_{f,g}$ is the del Pezzo surface of degree 6 or is obtained from it by toric blow-downs.*

Proof. The fan Σ of $X_{f,g}$ is smooth and complete. Its rays are among the six rays $\pm e_1$, $\pm e_2$, $\pm(e_1 + e_2)$ of the del Pezzo fan Σ_6. In cyclic order each ray of Σ_6 is the sum of its neighbours. Two of the six rays form a lattice basis exactly when they are at cyclic distance one or two; rays at distance three are opposite. So adjacent rays u, w of Σ are at distance one or two, and in the second case the omitted ray between them is $u + w$. Inserting $u + w$ is the star subdivision that blows up the corresponding fixed point [12]. Inserting all omitted rays turns Σ into Σ_6. □

Proposition 3.3 (Wall-crossing)**.** *On the wall $f_i = g_j$ three facet lines meet at a point. For example, on $f_1 = g_1$ the lines $x = g_1$, $y = 0$ and $x + y = f_1$ meet at $(g_1, 0)$. Crossing one wall cuts or restores one corner, which is one toric blow-up or blow-down. There are no flips, which do not occur for surfaces, and the type does not change inside a chamber.*

Proof. The concurrency is direct. For all 30 pairs of chambers whose sign vectors differ in one wall, the vertex counts differ by exactly one. Constancy inside a chamber is [2, Prop. 6.6]. □

Example 3.4.

(a) At $(f_1, g_1, g_2) = (0.1, 0.5, 0.2)$ the polygon is the triangle with vertices $(0,0)$, $(0.1,0)$ and $(0,0.1)$, and the surface is $\mathbb{CP}^2$.

(b) At $(0.4, 0.5, 0.3)$ the polygon is the pentagon with vertices $(0,0.2)$, $(0,0.3)$, $(0.1,0.3)$, $(0.4,0)$ and $(0.2,0)$, and the surface is $\mathrm{Bl}_2\mathbb{CP}^2$. The straight path to the uniform point crosses only the wall $f_1 = g_1$; the corner at $(f_1, 0)$ is cut and the hexagon appears.

(c) At the uniform point $(\frac{1}{2}, \frac{1}{3}, \frac{1}{3})$ the hexagon has vertices $(0, \frac{1}{6})$, $(0, \frac{1}{3})$, $(\frac{1}{6}, \frac{1}{3})$, $(\frac{1}{3}, \frac{1}{6})$, $(\frac{1}{3}, 0)$ and $(\frac{1}{6}, 0)$, all six edges have lattice length $\frac{1}{6}$, its lattice area is $\frac{1}{12}$, and $[\omega] = -\frac{1}{6}K_X$. This is the unique monotone point of the family [2, Prop. 6.7]: monotonicity holds at a point, not on a chamber.

Proposition 3.5 (Kähler data in the family)**.** *In all 18 chambers, $\omega_{f,g} \cdot D_c$ equals the lattice length of the edge $\{\alpha_c = 0\}$ and is positive, so $\omega_{f,g}$ is ample, and $\frac{1}{2}\omega_{f,g}^2$ equals the lattice area of* $\mathrm{Cpl}(f,g)$.

Proof. These are proposition 2.2 and the Duistermaat–Heckman formula for the volume. The script verifies both in exact arithmetic, using the intersection matrix of each fan. □

4 The square 2×2 case

Let $B = B' = \{0, 1\}$. A 1-cell is a stochastic matrix $k(a, b)$ with rows $(1 - a, a)$ and $(1 - b, b)$, and Kleisli composition is matrix multiplication. The deterministic 1-cells are the constants $e_0 = k(0, 0)$ and $e_1 = k(1, 1)$, the identity $\sigma = k(0, 1)$ and the swap $\tau = k(1, 0)$. A coupling 2-cell $\alpha\colon f \Rightarrow g$ is a coupling $\alpha(x)$ of $f(x)$ and $g(x)$ at each x. Its cost is $c(\alpha) = \max_x \Pr_{\alpha(x)}(y \neq y')$. Vertical composition is the gluing lemma, strictly associative with diagonal identities [2, Thm. 2.3].

Proposition 4.1 (Coupling segments and costs)**.** *For $f(x) = (p, 1 - p)$ and $g(x) = (q, 1 - q)$, the couplings are $\left(\begin{smallmatrix} a & p-a \\ q-a & 1-p-q+a \end{smallmatrix}\right)$ with $\max(0, p + q - 1) \leq a \leq \min(p, q)$. The product coupling has $a = pq$, and the maximal coupling, of cost $\mathrm{TV}(f(x), g(x))$, has $a = \min(p, q)$. Between deterministic 1-cells the coupling is unique, and its cost is 0 for equal 1-cells and 1 for any two distinct ones, because distinct deterministic maps differ at some x and the cost is a maximum over x.*

Definition 4.2 (Whiskering)**.** Precomposition by a deterministic $u\colon W \to B$ is $(\alpha \circ u)(w) = \alpha(u(w))$. Postcomposition by $h\colon B' \to \mathsf{D}(B'')$ copies h on equal pairs and runs it independently on unequal pairs:

$$(h \cdot \alpha)(x)(z, z') = \sum_y \alpha(x)(y, y)\, h(y)(z)\, [z = z'] + \sum_{y \neq y'} \alpha(x)(y, y')\, h(y)(z)\, h(y')(z').$$

This is the whiskering of [2, Thm. 2.3(ii)]. It sends diagonal couplings to diagonal couplings. Running h independently also on equal pairs would not preserve identities: for $h = k(\frac{9}{10}, \frac{1}{10})$ it sends $\frac{1}{2}\Delta$ to $\left(\begin{smallmatrix} 0.41 & 0.09 \\ 0.09 & 0.41 \end{smallmatrix}\right)$, so it cannot be the first component of any functor.

Proposition 4.3 (Whiskering defects)**.** *Let $F_2(\beta, \alpha) = (h \cdot \beta) \cdot (h \cdot \alpha) - h \cdot (\beta \cdot \alpha)$.*

(i) Precomposition by a deterministic map is strictly functorial.

(ii) If α is an identity, then $F_2(\beta, \alpha) = 0$.

(iii) With $f = g$ having uniform rows and $\alpha = \beta$ the swap coupling at each x, $F_2(\beta, \alpha)(x) = t\left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, where $t = \frac{369}{2500}$ for $h = k(\frac{9}{10}, \frac{1}{10})$ and $t = \frac{1}{4}$ for h constant and uniform.

(iv) Every defect has zero row and column sums. It is a tangent vector of the coupling polytope of $h^{\#} f(x)$ and $h^{\#} k(x)$, not a coupling.

Proof. (i) Both sides evaluate the same couplings at $u(w)$. (ii) $h \cdot \Delta = \Delta$, and Δ is a unit for gluing. (iii) $\beta \cdot \alpha = \mathrm{swap} \cdot \mathrm{swap} = \Delta$, so $h \cdot (\beta \cdot \alpha)$ is diagonal, while $h \cdot \mathrm{swap}$ has mass off the diagonal; the values are exact computations. (iv) Both terms couple the same two marginals. □

So the defect lives in the tangent space of a coupling polytope. There the Fisher metric, which is twice the Hessian of Guillemin's potential [2, Prop. 6.5], gives it a size. This is the only sense in which the defect meets the toric geometry that we can establish.

5 Second- and third-order states

Let $Y \subseteq \mathsf{D}B$ be a finite set of n first-order states and $P, Q \in \mathsf{D}(Y)$ second-order states with full support. A level-two coupling is $\Pi \in \mathsf{D}(Y \times Y)$ with marginals P and Q, and $\mathrm{Cpl}_2(P, Q)$ is the transportation polytope on $K_{n,n}$ with these marginals.

Proposition 5.1 (Delzant at levels two and three)**.** *If (P, Q) is nondegenerate, then $\mathrm{Cpl}_2(P, Q)$ is Delzant of dimension $(n-1)^2$. The same holds for third-order states $R, S \in \mathsf{D}(Z)$, with Z a finite set of second-order states.*

Proof. Theorem 2.1 uses only the marginals, not what the points of Y or Z are. □

Flattening. The map

$$\mu^*\colon \mathrm{Cpl}_2(P, Q) \to \mathrm{Cpl}(\mu_B P, \mu_B Q), \qquad \mu^*(\Pi)(b, b') = \sum_{\rho, \rho'} \Pi(\rho, \rho')\, \rho(b)\, \rho'(b'),$$

is affine. Its image contains $\mu_B P \otimes \mu_B Q$, the image of $P \otimes Q$.

Proposition 5.2 (Properties of flattening)**.**

(i) If Y consists of point masses, μ^* is the identity.

(ii) μ^* is not surjective in general: for $Y = \{\rho\}$ with ρ uniform on two points the image is $\{\rho \otimes \rho\}$, while $\mathrm{Cpl}(\rho, \rho)$ is a segment containing the diagonal coupling.

(iii) For $n = 2$ and $\rho_1 \neq \rho_2$, μ^* is injective: the segment direction maps to $(\rho_1 - \rho_2) \otimes (\rho_1 - \rho_2) \neq 0$.

(iv) If $(n-1)^2 > (|B|-1)^2$, the fibres of μ^* are positive-dimensional. For example, with three states on two outcomes and uniform $P = Q$ the fibre has dimension 3, so level-two couplings carry information that level-one couplings lose.

(v) For a kernel $k\colon B \to \mathsf{D}(B')$, pushing Π forward along $k^\# \times k^\#$ commutes with flattening: $\mu^*\big((k^\# \times k^\#)_* \Pi\big) = (k \otimes k)^\# \mu^*(\Pi)$. This is the level-two form of stochastic change of base [5, Prop. 9.14].

Proof. (i) $\rho(b) = [\rho = \delta_b]$. (ii) and (iii) are direct. (iv) The image lies in a space of dimension $(|B|-1)^2$; the example is computed exactly. (v) Both sides equal $\sum \Pi(\rho, \rho')\, k^\# \rho \otimes k^\# \rho'$. □

Uniform marginals. Uniform $n \times n$ marginals are degenerate for every $n \geq 2$. For $n = 2$ the polytope is still a segment, so $X = \mathbb{CP}^1$ and it is monotone, with special point the product coupling. For $n \geq 3$ the polytope is not simple and has no smooth toric manifold.

6 Disc potentials and optimal transport

For a Delzant polytope with facet functions ℓ_c and normals v_c, the potential of the torus fibre over u is $W_u(z) = \sum_c T^{\ell_c(u)} z^{v_c}$ over facets, for compact toric Fano manifolds [8] and, with corrections, for general compact toric manifolds [11]. A fibre with a flat connection has nonvanishing Floer cohomology when it corresponds to a critical point. For coupling polytopes the facets are cells, so W has at most nm terms, with exponents given by the normals of section 2. The facets are not the cycles of $K_{n,m}$.

Proposition 6.1 (The del Pezzo potential)**.** *At the uniform 2×3 point all facet functions equal $\frac{1}{6}$ at the product coupling, and $W = T^{1/6}(x + y + x^{-1} + y^{-1} + xy + x^{-1}y^{-1})$. Up to the factor $T^{1/6}$, W has six critical points: $(1, 1)$, $(1, -1)$, $(-1, 1)$, $(-1, -1)$, (ζ, ζ) and (ζ^2, ζ^2) for a primitive cube root of unity ζ. All six have $|x| = |y| = 1$, so they lie over the product coupling and differ only in the flat connection. The critical values are $6, -2, -2, -2, -3, -3$.*

Proof. Exact solution of $x\partial_x W = y\partial_y W = 0$; for instance $W(\zeta, \zeta) = 3\zeta + 3\zeta^2 = -3$. The number 6 equals twice the area 3 of the Newton hexagon, as Kushnirenko's theorem predicts. □

Example 6.2 (Non-monotone cases)**.**

(a) *A segment.* For the 2×2 segment $[\max(0, p+q-1), \min(p, q)]$, $X = \mathbb{CP}^1$ and the special fibre is the midpoint. That is the product coupling $a = pq$ when $p = \frac{1}{2}$, but not in general: for $p = q = \frac{1}{3}$ the midpoint is $\frac{1}{6}$ and $pq = \frac{1}{9}$.

(b) *A 3×3 case.* For $P = (0.40, 0.35, 0.25)$ and $Q = (0.30, 0.43, 0.27)$ the polytope is Delzant with 18 vertices and 9 facets, one per cell, so W has 9 terms.

Proposition 6.3 (Square cases are not Fano)**.** *Let $n \geq 3$ and let (f, g) be a nondegenerate $n \times n$ pair for which every cell is a facet. Then $X_{f,g}$ is not Fano. In the 3×3 example above, $-K_X$ is nef and big, so X is weak Fano.*

Proof. $-K_X = \sum_c D_c$ corresponds to the polytope $\{\langle u, v_c \rangle \geq -1 \ \forall c\}$, which is the transportation polytope with all row and column sums n, that is n times the Birkhoff polytope. If $-K_X$ were ample, this polytope would have normal fan equal to the smooth fan of X and would be simple, but the Birkhoff polytope is not simple for $n \geq 3$. For nefness, each maximal cone (a spanning tree T) must give a vertex of that polytope, that is, the tree solution of T with all marginals equal to n must be nonnegative. This is checked for all 18 trees of the example. Bigness is full-dimensionality. □

So Cho–Oh's Fano theorem applies to the 2×3 family, whose surfaces are all Fano, but not to square cases with $n \geq 3$, where the general theory of [11] is needed.

Proposition 6.4 (Optimal transport is a moment-map problem)**.** *For a cost $c \colon Y \times Y' \to \mathbb{R}$, the optimal-transport value $\min_{\alpha \in \mathrm{Cpl}(f,g)} \langle c, \alpha \rangle$ is the minimum over $X_{f,g}$ of the function $\langle c, \mu \rangle$, where μ is the moment map. It is attained at a torus-fixed point, that is, a vertex: a coupling supported on a spanning tree. If c is integral on Λ, then $\langle c, \mu \rangle$ is the Hamiltonian of a circle action.*

Proof. μ maps $X_{f,g}$ onto $\mathrm{Cpl}(f, g)$. A linear function on a polytope is minimized at a vertex, and the preimage of a vertex is a fixed point. A component $\langle \xi, \mu \rangle$ with ξ in the lattice generates the circle action of ξ [10]. The script checks the first statement by linear programming against vertex enumeration. □

This is where the polytope picture meets the obstruction theory of [5]. Kantorovich comparisons between second-order states [5, Prop. 9.14], and distances to effective sets, are linear programs over polytopes of this kind. Their optima are spanning-tree couplings, and their dual variables are the certificates. These vertices are not "committed" states in the tower's sense. A spanning-tree coupling has $n + m - 1$ nonzero entries and is not a point mass.

7 Stability conditions

A Bridgeland stability condition on a triangulated category is a pair $(Z, \mathcal{A})$: the heart $\mathcal{A}$ of a bounded t-structure, and an additive central charge $Z\colon K_0 \to \mathbb{C}$ that maps nonzero objects of $\mathcal{A}$ into the upper half-plane together with $\mathbb{R}_{<0}$, with Harder–Narasimhan filtrations and the support property [6]. The stability conditions form a complex manifold [6]. On any smooth projective surface, for any ample class ω, there is one with central charge $Z_\omega(E) = -\int e^{-i\omega}\operatorname{ch}(E)$ and a tilted heart [1, 19]. (Bridgeland's earlier construction [7] is for K3 surfaces.)

Proposition 7.1 (Marginals give stability conditions)**.** *For (f, g) in a chamber of the 2×3 family, $\omega_{f,g}$ defines a stability condition $\sigma_{f,g}$ on $D^b(\operatorname{Coh} X_{f,g})$, depending continuously on (f, g) within the chamber, with*

$$Z(\mathcal{O}_X) = \operatorname{area}\operatorname{Cpl}(f, g), \qquad Z(\mathcal{O}_x) = -1, \qquad Z(\mathcal{O}_{D_c}) = \tfrac{1}{2}D_c^2 + i\ \operatorname{length}\{\alpha_c = 0\}.$$

As (f, g) approaches a wall from inside a chamber, one edge length tends to 0, its divisor E is a (-1)-curve, and $Z(\mathcal{O}_E) \to -\frac{1}{2}$.

Proof. $\omega_{f,g}$ is ample by proposition 3.5. The values follow from $\operatorname{ch}(\mathcal{O}_X) = 1$, $\operatorname{ch}(\mathcal{O}_x) = [\mathrm{pt}]$ and $\operatorname{ch}(\mathcal{O}_D) = D - \frac{1}{2}D^2$, together with proposition 3.5. The statement about walls is proposition 3.3 with the self-intersections of theorem 3.1, and is checked in [2, §8]. □

Mirror symmetry. For a toric Fano manifold the disc potential is the Hori–Vafa Landau–Ginzburg potential [8]. The disc potential enters central charges through oscillatory integrals $\int e^{-W/z}$ over integral cycles, whose asymptotics involve the $\widehat{\Gamma}$-class [17, 13], not through its values. The critical values in proposition 6.1 are the eigenvalues of quantum multiplication by c_1, which is what the Gamma conjectures concern.

Question 7.2.

(i) Does the path $\sigma_{f,g}$ continue across a wall, glued along Orlov's decomposition $D^b(\mathrm{Bl}_p X) = \langle \mathcal{O}_E(-1), D^b(X)\rangle$ [20]?

(ii) Is there a natural object attached to an interior coupling, for instance the mirror of its torus fibre with a flat connection, whose central charge has a probabilistic meaning?

(iii) Do the quantum-corrected central charges with $\omega = \omega_{f,g}$ define stability conditions on these surfaces, and does the monotone point play a special role?

Not claimed. We make no identification of the disc potential with a central charge (it is a function on a torus, not an additive map on K_0), of committed states, the spanning-tree couplings, with a heart (they form no abelian category), of the obstruction tower with Harder–Narasimhan filtrations (the tower assigns numbers to a claim and filters no object), or of hallucination with instability. What is claimed instead is proposition 7.1 and, for hallucination, obstructed lifting in the tower [2, §11]. Each item is tabulated with its reason in [2, App. A].

8 The A_∞ question

The hom-categories of coupling 2-cells are strictly associative, so as A_∞-categories they have $m_k = 0$ for $k \geq 3$. If a linearization also has $m_1 = 0$, the arity-two A_∞-functor equation reduces to

$F_1(m_2(\beta,\alpha)) = m_2(F_1\beta, F_1\alpha)$, which says that whiskering is functorial. Proposition 4.3(iii) shows that it is not. Hence the defect can be the second component of an A_∞-functor only in a linearization with nonzero m_1, in which the defects are boundaries. The square-zero cone of the circulation ideal is such a linearization: into it, whiskering is an A_∞-functor with exactly two components (proposition 9.4; [2, Thm. 5.2]). Larger models such as the free cone remain open [2, Question 9.4]. Statements identifying the defect with the Hessian of a disc potential, or the functor equation with a Picard–Lefschetz formula, are withdrawn (appendix A). Section 9 gives the details.

9 Cumulants and the homotopy viewpoint

Homotopy probability theory treats an expectation that fails to be an algebra map as a morphism of homotopy algebras. Its components are then the cumulants [15, 16]. This section shows that the coupling polytopes carry exactly this structure. Couplings are second mixed cumulants, level-two flattening keeps only second cumulants, and the whiskering defects are the Boolean cumulants of the whiskering map. All of them are tangent vectors of coupling polytopes. Everything here is checked in `check_cumulants.py`.

Proposition 9.1 (Couplings are second cumulants)**.** *For $\alpha \in \mathrm{Cpl}(f,g)$, the centred coupling $\alpha - f\otimes g$ is the cross-covariance matrix $\kappa_2(1_{\{y=i\}}, 1_{\{y'=j\}})$ of the indicator vectors under α. So $\mathrm{Cpl}(f,g) - f\otimes g$ is the set of cross-covariances compatible with the marginals and with positivity. It lies in the space of circulations, and the product coupling is the point where every mixed cumulant of the pair (y,y') vanishes. The product coupling is also the only point of $\mathrm{Cpl}(f,g)$ of rank one, that is, the only point on the toric variety of the independence model. This is the Birch point of that model [21].*

Proof. $\mathbb{E}[1_{\{y=i\}}1_{\{y'=j\}}] - \mathbb{E}[1_{\{y=i\}}]\mathbb{E}[1_{\{y'=j\}}] = \alpha_{ij} - f_i g_j$. A nonnegative rank-one matrix $uv^\top$ with row sums f and column sums g equals $f\otimes g$. For two finite variables, independence is equivalent to vanishing of all mixed cumulants of the indicators. □

Proposition 9.2 (Level two: flattening keeps the second cumulant)**.** *For a level-two coupling $\Pi \in \mathrm{Cpl}_2(P,Q)$,*

$$\mu^*(\Pi) = \mu_B P \otimes \mu_B Q + \mathrm{Cov}_\Pi(\rho,\rho'),$$

where $\mathrm{Cov}_\Pi(\rho,\rho') = \mathbb{E}_\Pi[(\rho-\mu_B P)\otimes(\rho'-\mu_B Q)]$ is the cross-covariance of the two random measures, a circulation. The fibres of μ^* are therefore the level-two couplings with a fixed cross-covariance. For three states on two outcomes, the mixed moments $\mathbb{E}_\Pi[\rho(0)^a\rho'(0)^b]$ with $a,b\le 2$ determine Π, while flattening sees only $a=b=1$; this accounts for the three-dimensional fibres of proposition 5.2(iv).

Proof. Expand $\mathbb{E}_\Pi[\rho\otimes\rho']$. For the example, the map from the four-dimensional space of circulations to the mixed moments of order up to $(2,2)$ has rank 4, and to the order-$(1,1)$ moment it has rank 1. □

So level-two couplings carry the higher mixed cumulants of a pair of random distributions, and level-one gluing keeps only the second. This is the cumulant form of "level-two gluing keeps correlations that level-one gluing loses" [3, Prop. 7.16].

Whiskering as an expectation. Fix $f \in \mathsf{D}(Y)$ and let A_f be the space of real $Y\times Y$ matrices whose row sums and column sums are both multiples of f; it is spanned by the couplings of f with itself. With the gluing product $a\cdot b = a\,\mathrm{diag}(f)^{-1}b$ and unit $\mathrm{diag}(f)$, A_f is an associative unital algebra. The circulations I_f form a two-sided ideal, and $A_f/I_f \cong \mathbb{R}$ records only the total mass.

Whiskering by a kernel h (definition 4.2) is a linear unital map $h_*\colon A_f \to A_{h\#f}$, a noncommutative "expectation" that need not be multiplicative. Its *Boolean cumulants* [22] are defined recursively over interval partitions:

$$b_1 = h_*, \qquad b_2(a,b) = h_*(a\cdot b) - h_*(a)\cdot h_*(b),$$
$$b_3(a,b,c) = h_*(a\cdot b\cdot c) - h_*a\cdot h_*b\cdot h_*c - b_2(a,b)\cdot h_*c - h_*a\cdot b_2(b,c).$$

Here $a\cdot b$ means: glue a first, then b.

Proposition 9.3 (Whiskering defects are Boolean cumulants)**.**

(i) The whiskering defect of proposition 4.3 is $F_2(\beta,\alpha) = -b_2(\alpha,\beta)$.

(ii) Every b_n with $n \geq 2$ takes values in the circulations $I_{h\#f}$, that is, in tangent vectors of coupling polytopes. Modulo $I_{h\#f}$, h_* is an algebra map.

(iii) For $|Y| = 3$ the defects b_2 span all of $I_{h\#f}$, and b_3 is nonzero in general.

(iv) Let C be the differential graded algebra obtained from $A_{h\#f}$ by freely adjoining generators $\delta(a,b)$ of degree -1 with $d\delta(a,b) = b_2(a,b)$; this is a cone in the sense of homotopy probability theory. Then (h_*,δ) satisfies the arity-two equation of an A_∞-functor $A_f \to C$, the arity-three obstruction is a cycle, and $H^0(C) = A_{h\#f}/(b_2)$, which equals $A_{h\#f}/I_{h\#f} \cong \mathbb{R}$ whenever the defects span $I_{h\#f}$, as in (iii).

Proof. (i) is the definition. (ii) Both composites couple the same two marginals, so their difference is a circulation; the recursion preserves this because I is an ideal. Modulo I only masses remain, and they multiply. (iii) The defects from 40 random instances have rank $4 = \dim I$ for three points, and random instances give $b_3 \neq 0$. (iv) The arity-two equation is $d\delta = b_2$. The arity-three obstruction is $h_*a\cdot\delta(b,c) - \delta(a,b)\cdot h_*c - \delta(a\cdot b,c) + \delta(a,b\cdot c)$, whose differential vanishes identically; this is verified exactly. H^0 of a free extension by degree -1 generators is the quotient by the two-sided ideal generated by their boundaries, here $I_{h\#f}$ by (iii). □

Proposition 9.4 (A two-component A_∞-functor into the square-zero cone)**.** *Let $C = A_{h\#f}\oplus\varepsilon I_{h\#f}$, where ε has degree -1, $\varepsilon^2 = 0$, $d(\varepsilon x) = x$, and $A_{h\#f}$ acts on εI through the ideal structure. Then C is a differential graded algebra with $H^0(C) = A_{h\#f}/I_{h\#f}$ and $H^{-1}(C) = 0$, so it is the mapping cone of $I \hookrightarrow A$, a model of the marginal quotient. Whiskering extends to an A_∞-functor $A_f \to C$ with*

$$F_1 = h_*, \qquad F_2(a,b) = \varepsilon\, b_2(a,b), \qquad F_n = 0 \quad (n \geq 3).$$

For three and four points the defects span I, with rank $(n-1)^2$.

Proof. C is the trivial extension of A by the bimodule εI; d is a derivation because I is an ideal, and d is injective on εI. In arity two the functor equation is $dF_2(a,b) = h_*(ab) - h_*a\cdot h_*b$, which holds by definition. In arity three, with $F_3 = 0$ (there is nothing in degree -2), it reads $b_2(ab,c) - b_2(a,bc) = h_*a\cdot b_2(b,c) - b_2(a,b)\cdot h_*c$. Expanding with $b_2(x,y) = h_*(xy) - h_*x\,h_*y$, the $h_*(abc)$ terms cancel on the left and the $h_*a\,h_*b\,h_*c$ terms cancel on the right, and both sides become $h_*a\cdot h_*(bc) - h_*(ab)\cdot h_*c$. Only linearity of h_* and associativity of gluing are used, so this is a proof for all sizes [2, Thm. 5.2]; b_2 is the Hochschild coboundary of h_*. In arity 4 the only surviving term is $F_2\cdot F_2 \in \varepsilon^2 I = 0$, and in arity $n \geq 5$ every term contains some F_k with $k \geq 3$. The script runs the identity on random inputs as a unit test, and computes the ranks for $n = 3, 4$, which are a computation. □

So the answer to the first part of the question below is that no higher components are needed in this model. The higher Boolean cumulants b_n, which are nonzero, appear only in larger models such as the free cone. This parallels homotopy probability theory, where a cone factors the expectation as an inclusion followed by a quasi-isomorphism [15].

What this shows. Making whiskering an A_∞-functor through a cone forgets exactly the correlations. The cohomology H^0 keeps only the marginals, that is, the Kleisli level of [2]. The correlation structure lives in the homotopy data: the Boolean cumulants b_n, which are tangent vectors of the coupling polytopes. In the geometry of section 2, the base of the fibration $\mathrm{Cpl}(f, g) \mapsto (f, g)$ is the cohomological part, and the fibres, where the Fisher metric measures the cumulants [2, Prop. 6.5], carry the homotopy. This also explains section 8: with $m_1 = 0$ the cumulants have nowhere to live, and a cone supplies the missing m_1.

Question 9.5.

(i) Into the square-zero cone, whiskering is an A_∞-functor with two components (proposition 9.4). Into the free cone, do the higher Boolean cumulants b_n, with generators of degree $1 - n$, give the higher components, as in the bar-construction description of Boolean cumulants?

(ii) Classical and free cumulants are organized by set partitions and non-crossing partitions, and A_∞-structures by associahedra. The permutohedron is a linear image of the Birkhoff polytope, and its faces correspond to ordered set partitions. Uniform square coupling polytopes are scaled Birkhoff polytopes, which lie on the walls of the chamber structure. Does cumulant combinatorics live on these degenerate walls?

10 The coupling nerve

The theory paper [2] organizes couplings into a simplicial object. This section computes its low-dimensional pieces. The *coupling nerve* on a finite set Y is

$$N_k = \mathsf{D}(Y^{k+1}),$$

with face maps d_i that forget the ith coordinate (marginalization) and degeneracies s_i that repeat it (pushforward along the diagonal). It is D applied levelwise to the contractible Kan complex $Y^{\bullet+1}$. The relative version over a 1-cell, with kernels $X \to \mathsf{D}(Y_0 \times \cdots \times Y_k)$, works pointwise in x. A 1-simplex is a coupling, and a 2-simplex with faces $\alpha = d_2$, $\beta = d_0$ is a joint law whose d_1 face is a composite of α and β. Every claim below is checked in `check_nerve.py`.

10.1 2-horns: fillers form toric polytopes

Proposition 10.1 (2-horn fibres)**.** *Let p_{01} and p_{12} be compatible couplings on $Y_0 \times Y_1$ and $Y_1 \times Y_2$, with common marginal p_1. The fillers, meaning laws p on $Y_0 \times Y_1 \times Y_2$ with these two faces, form the product over x_1 of the transportation polytopes of the slices $p(\cdot, x_1, \cdot)$, which have marginals $p_{01}(\cdot, x_1)$ and $p_{12}(x_1, \cdot)$. The dimension is $|Y_1|(|Y_0| - 1)(|Y_2| - 1)$, and the fibre is Delzant when every slice is nondegenerate: a product of the toric manifolds of section 2.*

Proof. Every marginal constraint involves a single value of x_1, so the constraints decouple. The script checks the decoupling and the dimension for shapes $2 \times 2 \times 2$, $2 \times 3 \times 3$ and $3 \times 2 \times 3$, and Delzant is theorem 2.1 for each slice. □

Proposition 10.2 (The canonical filler)**.** *The gluing filler*

$$p^{\mathrm{gl}}(x_0, x_1, x_2) = p_{01}(x_0, x_1)\, p_{12}(x_1, x_2)/p_1(x_1)$$

is the unique filler of maximal entropy. Its d_1 face is the vertical composite of the two couplings. Any filler differs from it slice by slice by $p_1(x_1)\,\mathrm{Cov}(1_{\{x_0\}}, 1_{\{x_2\}} \mid x_1)$, a circulation.

Proof. $\log p^{\mathrm{gl}}$ is a sum of a function of (x_0, x_1) and a function of (x_1, x_2), so it is orthogonal to every tangent vector of the fibre. Hence p^{gl} is a critical point of the strictly concave entropy on the fibre, and so its unique maximizer. This is the inequality $H(X_0X_1X_2) \le H(X_0X_1) + H(X_1X_2) - H(X_1)$, with equality exactly under conditional independence. The covariance identity is direct. The script checks orthogonality to a basis of the tangent space, a strict decrease of entropy along random feasible directions, and the identity in exact arithmetic. □

So composition of 2-cells is the canonical (maximum-entropy) filling of 2-horns, and the other fillers are the same cumulant directions as in section 9, now taken conditionally on the middle vertex. This is the 2-horn analogue of the Birch point of proposition 9.1.

10.2 Higher horns do not always fill

Proposition 10.3 (Parity horns)**.** *On bits, let a horn Λ^n_k carry, on each of its n faces, the uniform law on assignments of fixed parity, with exactly one face odd. These data agree on all overlaps. For $n = 2$ every such horn fills. For every $n = 3$ horn the contextual fraction of the horn, computed as the minimal unmodeled weight with global assignments as sources [3, Thm. 7.12], is $\frac{1}{2}$. For every $n = 4$ horn it is $\frac{3}{4}$. Both values come with exactly verified certificates. The best approximate fillers are at distance $\varepsilon^* = \frac{1}{6}$ for $n = 3$ and $\varepsilon^* = \frac{3}{16}$ for $n = 4$. For all $n \ge 3$ the contextual fraction is $1 - 2^{2-n}$ [2, Thm. 4.3]; section 13.1 has the values up to $n = 8$.*

Corollary 10.4. *The coupling nerve $\mathsf{D}(Y^{\bullet+1})$ is not a Kan complex, although $Y^{\bullet+1}$ is. All 2-horns fill, and the obstruction to filling higher horns is measured by the contextual fraction. This matches Vorob'ev's theorem [23]: every compatible family on a cover extends exactly when the cover is acyclic, which holds for the two faces of a 2-horn and fails for the faces of an n-horn with $n \ge 3$.*

Compatible 3-horns that are mixtures $t \cdot (\text{parity}) + (1 - t) \cdot (\text{restriction of a global law})$ have contextual fraction at most $t/2$. In all 40 random samples the inequality was strict.

10.3 Boundary fibres: where the toric picture stops

Filling the whole boundary $\partial\Delta^2$ means fixing all three 2-margins of a 3-way table. The fibre is a planar 3-way transportation polytope of dimension $(n_0 - 1)(n_1 - 1)(n_2 - 1)$.

Proposition 10.5 (Boundary fibres)**.**

(i) For shape $2 \times n \times m$, the table is determined by its first layer $y = x(0, \cdot, \cdot)$, subject to fixed row and column sums and $0 \le y \le m_{12}$. So the fibre is a capacitated transportation polytope. Its constraint matrix is totally unimodular, so it is Delzant whenever it is simple. All generic samples of shape $2 \times 2 \times 2$, $2 \times 2 \times 3$ and $2 \times 3 \times 3$ were simple and smooth.

(ii) For shape $3 \times 3 \times 3$ (exploratory), generic samples are simple but not *smooth. In one instance with 1751 vertices, 202 vertices have edge vectors with entries ± 2 whose lattice has index 64 (Smith invariants $1, 1, 2, 2, 2, 2, 2, 2$). The fibre is therefore the moment polytope of a toric orbifold [18], not of a manifold.*

(iii) By De Loera and Onn, every rational polytope is a planar $r \times s \times 3$ transportation polytope [9]. So boundary fibres eventually realize arbitrary polytopes, and no toric regularity can hold for them in general.

Proof. (i) The (j,k)-margins give $x(1,j,k) = m_{jk} - y_{jk}$, and the other margins become the row and column sums of y. The constraint matrix is a bipartite incidence matrix stacked with an identity, which is totally unimodular, so edges are ± 1 cycle vectors, as in theorem 2.1. (ii) Vertices are enumerated exactly by simplex pivoting, adjacency is read off from zero patterns, and lattice indices are computed by Smith normal form. (iii) is the cited theorem. □

Filling problem	Fibre	Toric status
1-simplex over its two vertices	coupling polytope	Delzant if nondegenerate (theorem 2.1)
2-horn Λ^2_k	product of coupling polytopes	Delzant if slices nondegenerate (proposition 10.1)
$\partial\Delta^2$, shape $2 \times n \times m$	capacitated transportation	Delzant if simple (proposition 10.5)
$\partial\Delta^2$, shape $3 \times 3 \times 3$	planar 3-way table	simple, not smooth: an orbifold
$\partial\Delta^2$, general	planar 3-way table	arbitrary rational polytopes [9]
Λ^n_k, $n \geq 3$	may be empty	obstruction = contextual fraction (proposition 10.3)

11 The tower as a fibration: a worked example

The base-relative tower of [5] assigns to each finite set B a context algebra A_B, a reader comonad C^B and a strict mixed distributive law λ^B. Every Markov kernel $k\colon B \to \mathsf{D}(B')$ acts by $A_k = k^{\#} \times \mathsf{D}(k^{\#}) \times \mathsf{D}^2(k^{\#}) \times \mathsf{D}(\mathrm{Hyp}\, k)$, with $A_{h \circ k} = A_h A_k$ [5, Prop. 9.14]. So the tower is a functor on the Kleisli category, and its Grothendieck construction is a split opfibration over $\mathrm{Kl}(\mathsf{D})$. This section follows one explicit fibre along explicit kernels. Every number below is checked in `check_tower_fibration.py`, in exact arithmetic except for the linear programs.

The data. Take $B = \{0,1\}$, the posteriors δ_0, δ_1 and $\eta = (\frac{1}{2}, \frac{1}{2})$, and the general (non-saturated) admissibility data

$$S_1 = \{\delta_0, \delta_1, \eta\}, \qquad S_2 = \left\{\tfrac{1}{2}\delta_{\delta_0} + \tfrac{1}{2}\delta_{\delta_1},\ \delta_\eta\right\}$$

of [5, Def. 10.11]. A fair coin is admissible only as certainty in η or as a mixture of the two decisive posteriors. Then L_1 is the whole simplex, but the effective set is a single point, $E_1 = \{\eta\}$. The kernels are

$$k\colon\ 0 \mapsto (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}),\ 1 \mapsto (\tfrac{1}{5}, \tfrac{1}{5}, \tfrac{3}{5}) \quad \text{on } B' = \{u, v, w\}, \qquad \kappa\colon\ 0, 1 \mapsto (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}),$$

together with $h\colon B' \to \mathsf{D}\{P, Q\}$, given by $u \mapsto \delta_P$, $v \mapsto (\frac{1}{2}, \frac{1}{2})$, $w \mapsto \delta_Q$.

Statement	Values in the example
A claim passes the local test but has no lift	$x_1 = (\frac{7}{10}, \frac{3}{10}) \in L_1 \setminus E_1$; certificate $f = (1, -1)$ with $\langle f, x_1\rangle = \frac{2}{5} > 0 = \langle f, \eta\rangle$
Along k the failure persists, and certificates pull back	$k^{\#}x_1 = (\frac{41}{100}, \frac{47}{200}, \frac{71}{200}) \neq k^{\#}\eta = (\frac{7}{20}, \frac{9}{40}, \frac{17}{40})$; $f' = k^{\#}x_1 - k^{\#}\eta$ separates them, and $k \triangleright f' = (\frac{3}{200}, -\frac{7}{250})$ separates x_1 from E_1 on B
Along κ the failure disappears	$\kappa^{\#}x_1 = \kappa^{\#}\eta$: the failure level rises to ∞
The secondary obstruction can only fall	$o_3(\delta_{\delta_0}) = 1$ on B, 1 after k (injective on S_1), 0 after κ
Coherence and functoriality	a coherent context stays coherent under A_k, and $A_{h\circ k} = A_h A_k$ exactly
Grades transport to every level	for $g\colon 0 \mapsto (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $1 \mapsto (\frac{1}{10}, \frac{3}{10}, \frac{3}{5})$: $d(k, g) = \frac{1}{6}$, while the level-1 total variation and the level-2 and level-3 Kantorovich distances between $A_k(z)$ and $A_g(z)$ for a coherent context z are all $\frac{2}{15}$
Conditioning through the channel	observing w on B' after k equals conditioning on $k \triangleright 1_w$ on B, which has likelihoods $\frac{1}{4}, \frac{3}{5}, \frac{17}{40}$ on δ_0, δ_1, η; the posterior weights become $\frac{5}{34}, \frac{6}{17}, \frac{1}{2}$

The example shows the three properties of the fibration that matter for provenance. First, transport never creates an obstruction, and it can destroy one: here κ forgets the outcome. Second, a certificate found downstream is a certificate upstream. Third, the quantitative comparison of two kernels bounds the comparison of the transported states at every level of the tower, so grades of the 2-category become grades on fibres.

The same fibre with a support policy. Enlarge the fibre from points to credal sets [2, Def. 3.5, Prop. 3.6]. Take the segment L from η to x_1 and the same kernels k and h; the transport of policies and floors is [5, Prop. 10.24, Thm. 10.26], and this is its [5, Ex. 10.27].

Statement	Values in the example
Transport carries the set	$k^{\#}L$ is the segment from $(\frac{7}{20}, \frac{9}{40}, \frac{17}{40})$ to $(\frac{41}{100}, \frac{47}{200}, \frac{71}{200})$, with floor $(\frac{7}{20}, \frac{9}{40}, \frac{71}{200})$
Summarising loses certification, transport does not	carried as a set to $\{P, Q\}$, $\rho(Q) \geq \frac{189}{400}$; summarised to its floor box after k, only $\rho(Q) \geq \frac{187}{400}$: a strict oplax comparison
Policies before transport certify more	with $\rho(0) \geq \frac{3}{5}$ imposed before k the floor is $(\frac{19}{50}, \frac{23}{100}, \frac{71}{200})$; pushing the floor $k^{\#}\tau = (\frac{3}{10}, \frac{3}{20}, \frac{3}{20})$ instead certifies $(\frac{7}{20}, \frac{9}{40}, \frac{71}{200})$

A fourth property of the fibration for provenance follows: certification is never lost along a transport, only where a set is summarised by its floors, so the weakest link of a chain is a summary step. On the polytope side, a floor on one cell of the hexagon of uniform 2×3 margins is a reduced-margin polytope and tears it into a quadrilateral at $\tau = \frac{1}{6}$ [2, Prop. 6.9, Ex. 6.10]. Both are checked in exact arithmetic in `check_support_floors.py`.

12 The family over the space of marginals

The coupling polytopes form a family $\mathrm{Cpl}(f, g) \to (f, g)$ over the space of marginal pairs. This section records how the family varies, in the 2×3 case. The checks are in `check_family.py` and `check_family_geometry.py`.

Proposition 12.1 (Variation inside and across chambers)**.**

(i) Inside a chamber. *Each vertex is an affine function of* (f_1, g_1, g_2)*, given by the solution on a fixed spanning tree, and the set of trees is constant on the chamber. So the polygons in one chamber are a linear family with a fixed normal fan, and the edge lengths, hence the degrees* $\omega_{f,g} \cdot D_c$*, are affine in the marginals. In the hexagon chamber, with* $c = f_1 + g_1 + g_2 - 1$ *and coordinates* $(x, y) = (\alpha_{11}, \alpha_{12})$*, the vertices are* $(0, c)$*,* $(0, g_2)$*,* $(f_1 - g_2, g_2)$*,* $(g_1, f_1 - g_1)$*,* $(g_1, 0)$ *and* $(c, 0)$*.*

(ii) The product coupling $f \otimes g$ *is a polynomial section of the family that lies in the interior of every fibre. It is the maximum-entropy point of each fibre [2, Prop. 6.5].*

(iii) Across a wall. *Along the straight path from* $(0.4, 0.5, 0.3)$ *to the uniform point, which crosses exactly the wall* $f_1 = g_1$ *at* $t = \frac{3}{8}$*, the lattice area of the polygon is*

$$\tfrac{11}{200} + \tfrac{t}{25} + \tfrac{t^2}{450} \quad (t \le \tfrac{3}{8}), \qquad \tfrac{1}{20} + \tfrac{t}{15} - \tfrac{t^2}{30} \quad (t \ge \tfrac{3}{8}).$$

It is continuous and continuously differentiable at the wall, and its second derivative jumps by $\frac{16}{225}$*, which is the second derivative of the area* $\frac{1}{2}(f_1 - g_1)^2$ *of the corner that is cut off. The edge on* $\{\alpha_{10} = 0\}$ *appears at the wall with length* 0 *and grows linearly, as* $f_1 - g_1$*.*

(iv) Information geometry is blind to the walls. *The Fisher metric at the product coupling,* $G(v, w) = \sum_c v_c w_c / (f_i g_j)$ *on circulations, is a rational function of* t *with no singularity on the path. At the wall it equals* $\left(\begin{smallmatrix} 11264/441 & 1024/63 \\ 1024/63 & 1024/35 \end{smallmatrix}\right)$ *in the basis* e_{11}, e_{12}*.*

Proof. (i) is [2, Prop. 6.6] made explicit: tree solutions are linear in the marginals. The script checks affine interpolation between 189 pairs of points in common chambers. (ii) The entries $f_i g_j$ are positive. (iii) Exact second divided differences on each side, and one-sided difference quotients at the wall. The area is the Duistermaat–Heckman volume $\frac{1}{2}\omega_{f,g}^2$ (proposition 3.5), which is piecewise polynomial in the Kähler class. Cutting a corner of size s removes area of order s^2, which explains the C^1 behaviour. (iv) The product coupling never meets the boundary. □

t	vertices	length of $\{\alpha_{10} = 0\}$	area
0	5	0	11/200
1/4	5	0	469/7200
3/8 (wall)	5	0	9/128
1/2	6	1/30	3/40
3/4	6	1/10	13/160
1	6	1/6	1/12

So the two geometries on the family see different things. The toric and symplectic data (combinatorial type, edge lengths, Duistermaat–Heckman volume, the stability conditions of section 7) change at the walls, where an edge, and so a (-1)-curve, shrinks to zero. The information geometry at the product coupling varies smoothly through them. Walls in the space of marginals are therefore a toric phenomenon, not a statistical one: they mark where the set of possible couplings changes shape, not where the canonical coupling changes behaviour.

13 Computations for the five open problems

The theory paper lists five open problems and gives, in its §10, the experimental definitions and the results for each [2, §10]. This section records the numbers behind them. The scripts are listed in appendix B.

13.1 Higher horns

Proposition 10.3 extends to all n: the contextual fraction of the parity n-horn is $1 - 2^{2-n}$ [2, Thm. 4.3]. The best approximate filler is a global law that minimizes the largest total-variation distance between its face marginals and the horn data. Its distance $\varepsilon^*(n)$ comes from a linear program:

n	3	4	5	6	7	8
contextual fraction $1 - 2^{2-n}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{7}{8}$	$\frac{15}{16}$	$\frac{31}{32}$	$\frac{63}{64}$
$\varepsilon^*(n)$	0.1667	0.1875	0.2250	0.2500	0.2589	0.2832

The contextual fractions for $n \le 6$ have exactly verified certificates. For $n \le 8$ the script also confirms the count that proves the formula: exactly two global assignments satisfy all face parities. The values $\varepsilon^*(n)$ increase and stay below $\frac{1}{2}$. The value for $n = 8$ is $\frac{145}{512}$; we claim no closed form.

13.2 Wall-crossing beyond 2×3

The hyperplanes spanned by the GIT weights $e_i \oplus e_j$ are the walls together with the facets of the weight cone [2, Prop. 10.5]:

shape	hyperplane flats	walls $\sum_I f = \sum_J g$	facets $f_i = 0$, $g_j = 0$
2×3	11	6	5
2×4	20	14	6
3×3	24	18	6

Single-wall crossings were classified by comparing normal fans, with rays the facet cells and maximal cones the zero sets at vertices. For 2×4, 40 random crossings gave 18 blow-ups or blow-downs (one ray changes) and 22 flips (same rays, different cones). For 3×3, 25 crossings gave 6 and 19. No other type occurred.

13.3 Across the wall $f_1 = g_1$

On the path of section 12, with $s = f_1 - g_1 = \frac{4t}{15} - \frac{1}{10}$, the pentagon and hexagon areas are

$$\mathrm{area}_-(t) = \tfrac{(2t+3)(2t+33)}{1800}, \qquad \mathrm{area}_+(t) = \tfrac{3+4t-2t^2}{60} = \mathrm{area}_-(t) - \tfrac{1}{2}s^2.$$

The jump in the second derivative is therefore $(4/15)^2 = \frac{16}{225}$, as in proposition 12.1. The glued central charge of [2, Def. 10.7] gives:

t	$\frac{3}{8}$	$\frac{1}{2}$	$\frac{3}{4}$	1
s	0	$\frac{1}{30}$	$\frac{1}{10}$	$\frac{1}{6}$
$Z(\mathcal{O}_X) = \mathrm{area}$	$\frac{9}{128}$	$\frac{3}{40}$	$\frac{13}{160}$	$\frac{1}{12}$
$Z(\mathcal{O}_E(-1))$	$\frac{1}{2}$	$\frac{1}{2} + \frac{i}{30}$	$\frac{1}{2} + \frac{i}{10}$	$\frac{1}{2} + \frac{i}{6}$
phase of $\mathcal{O}_E(-1)$	0	0.0212	0.0628	0.1024

On $\pi^* K_0(X_-)$, $Z_+ - Z_- = -\,\mathrm{rk} \cdot \frac{1}{2}s^2$ exactly. The numerical jump at $t = \frac{3}{8} \pm 10^{-8}$ is 3×10^{-9}. The Euler form confirms Orlov's decomposition on K_0.

13.4 The multi-object cone

All identities of [2, Thm. 10.2, Prop. 10.4] are checked exactly on objects of sizes $2, 3, 4, 2$, on kernels with $|X| = 2$, and on random 3-way and 4-way tables. Two actions of a kernel are compared. The whiskering of definition 4.2 (copy on equal pairs) is strictly unital. The product action $a \mapsto h^{\mathsf{T}} a h$ is not, already for $f = (\frac{1}{2}, \frac{1}{2})$ and $h = \left(\begin{smallmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{smallmatrix}\right)$:

$$h^{\mathsf{T}} \operatorname{diag}(f)\, h - \operatorname{diag}(\tfrac{3}{4}, \tfrac{1}{4}) = \tfrac{1}{8} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \in \mathcal{I}.$$

In exchange the product action composes strictly along kernels, while the whiskering does so only for deterministic kernels. For two random stochastic 3×3 kernels the two composites differ [2, Thm. 10.2(iv)].

13.5 The Fukaya probe

At the monotone point the six critical points of W lie over the product coupling, and the Hessian at $(1, 1)$ is $\frac{1}{6}$ of the Fisher metric. Away from it, the locations $u_k = -\lim \log|z_k|/\lambda$ of the six critical points are:

(f_1, g_1, g_2)	leximin (multiplicity 4)	other two	product coupling
$(0.52, 0.31, 0.35)$	$(0.155, 0.1875)$	$(0.17, 0.21)$, $(0.18, 0.13)$	$(0.1612, 0.182)$
$(0.47, 0.09, 0.46)$	$(0.045, 0.2125)$	$(0.01, 0.45)$, $(0.02, 0.02)$	$(0.0423, 0.2162)$
$(0.46, 0.45, 0.36)$	$(0.185, 0.18)$	$(0.44, 0.01)$, $(0.09, 0.27)$	$(0.207, 0.1656)$
$(0.53, 0.45, 0.23)$	$(0.255, 0.115)$	$(0.37, 0.08)$, $(0.02, 0.21)$	$(0.2385, 0.1219)$

Coordinates are $(\alpha_{00}, \alpha_{01})$. At all four sampled points the positive real critical point lies over the leximin coupling; this is computed, not proved [2, Obs. 10.10]. That the product coupling is absent is proved whenever its smallest cell is unique [2, Prop. 10.11], which is the case at all four points. At the first point the two other couplings are signed-balanced: the minimal cells are $\{02, 10, 11\}$ with holonomy $(-1, -1)$, and $\{01, 10, 12\}$ with holonomy $(-1, 1)$. At $\lambda = 40$ the cosine between the Hessian of $\mathfrak{PD}$ at the positive critical point and the Fisher metric at the product coupling is 0.9866, not 1.

14 The obstruction tower and the nerve: computations

This section gives the numbers behind [2, §11]. A horn on bits is encoded as a level-one claim on the disjoint union of its face assignment sets, with weight $\frac{1}{n}$ per face. Global assignments are the effective mechanisms. For the parity horns:

n	CF	$d(x_1, E_1)$	$2\varepsilon^*(n)$	o_3^-, $S_1 = \text{vert Loc}$	o_3^-, $S_1 = \text{global + parity}$
3	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	2
4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{2}$	2
5	$\frac{7}{8}$	$\frac{9}{20}$	$\frac{9}{20}$	$\frac{7}{4}$	2

Each row passes the local test, so it is extension-obstructed. The secondary obstruction equals twice the contextual fraction exactly when every compatible family is admissible. Otherwise it depends on which contextual mechanisms are admitted. The ℓ^1 distance to the fillable horns equals twice the best approximate filler distance of section 13.1. For mixtures of a parity horn with a restricted global law, all three quantities scale down together and vanish when the mixture is fillable; in every computed case $d(x_1, E_1) \le 2\,\mathrm{CF} \le o_3^-$. Every compatible 2-horn lies in E_1.

Directions. With only global mechanisms admissible, the fibre of level-two lifts of a 2-horn is its filler polytope. Its tangent space consists of the slice-wise circulations, of dimension $\sum_{y_1}(|Y_0|-1)(|Y_2|-1)$: 2, 3 and 12 for shapes $2\times2\times2$, $2\times3\times2$ and $3\times2\times4$.

Fisher bound. In 300 random instances, $d(x_1, E_1)^2 \le \|x_1 - y\|_1^2 \le \min\{\chi^2(x_1\|y), 2\,\mathrm{KL}(x_1\|y)\}$ for $y \in E_1$, with the largest ratio $\|x_1 - y\|_1^2/\chi^2$ equal to 0.9998. In the example of section 11, $x_1 = (\frac{7}{10}, \frac{3}{10})$ and $E_1 = \{h\}$, the bound is attained: $0.16 = 0.16$, because $|x_1 - h|$ is proportional to h.

Transport. In 60 random general admissibility data transported along random kernels, o_3^- never increased, and it decreased strictly in 12 cases. This matches the worked example of section 11, where o_3 stays at 1 under an injective kernel and drops to 0 under a constant one.

15 Synthetic data

This section tabulates the synthetic data behind [2, §12]. The data are generated by the seeded script `synthetic_examples.py` (appendix B), which writes each data set as a CSV file; the files are archived at https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/blob/main/coupling/synthetic_data.zip. The same script produces the deep-ensemble examples of [5, Section 11]. Real-data counterparts are in [4]: bounds over all gluings on media-bias corpora (Section 38.5 there), the sampling floor of the contextual fraction (Section 38.8), and obstructed claims that cannot learn (Section 52).

15.1 A confusion table is a coupling

The label Y is safe or harmful, and the action A is accept, review or reject. There are $N = 2000$ samples, in `triage_samples.csv`:

	true α			empirical $\hat{\alpha}$		
	accept	review	reject	accept	review	reject
safe	0.28	0.20	0.04	0.311	0.188	0.041
harmful	0.03	0.15	0.30	0.021	0.151	0.290

The marginals $f = (0.52, 0.48)$ and $g = (0.31, 0.35, 0.34)$ lie in the hexagon chamber of section 3. The wall values $\sum_I f - \sum_J g$ are $0.21, 0.17, 0.18, -0.14, -0.13, -0.17$, so the distance to the nearest wall is $\delta = 0.13$. From the marginals alone, the Fréchet interval for $P(\text{harmful}, \text{accept})$ is $[0, 0.31]$; the true value is 0.03.

15.2 Chamber identification

The table gives the observed probability that the empirical marginals land in the wrong chamber, over 4000 replications far from a wall and 2000 near one. The bound is $\min\{1, 12e^{-N\delta^2/2}\}$, from [2, Prop. 12.2]. The data are in `chamber_identification.csv`.

far from a wall, $\delta = 0.13$					
N	50	100	200	500	1000
error	0.303	0.085	0.013	0	0
bound	1	1	1	0.18	0.003
near a wall, $\delta = 0.02$, $g = (0.31, 0.19, 0.50)$					
N	10^2	10^3	$5 \cdot 10^3$	$2 \cdot 10^4$	$5 \cdot 10^4$
error	0.661	0.356	0.056	0	0
bound	1	1	1	0.22	0.0005

The bound is valid and conservative, by roughly a factor of three in N. Near a wall, about 5000 samples are needed before the surface is identified reliably.

15.3 Independence tests

For the empirical table, $X^2 = N\, g^{\mathrm{F}}(v, v) = 2000 \times 0.4418 = 883.7$, $G = 2N$ KL $= 2000 \times 2 \times 0.2567 = 1026.7$, and the estimated mutual information is 0.257 nats. Rejection rates at level 0.05 with critical value $\chi^2_{2,0.95} = 5.99$, so two degrees of freedom, the dimension of the hexagon (`independence_tests.csv`):

setting	N	replications	Pearson	G
null: the product coupling $f \otimes g$	500	4000	0.046	0.047
alternative: the true table	50	4000	0.997	0.998

15.4 Two models on one label set

A second model B has confusion table rows $(0.25, 0.22, 0.05)$ and $(0.05, 0.13, 0.30)$. Given the label, its errors are correlated with those of A: the joint law mixes conditional independence (40%) with the agreement-maximizing coupling (60%). The target is the probability that both models accept a harmful item (`two_models_samples.csv`, $N = 5000$):

	sharp bounds from the two tables	gluing (conditionally independent)	truth
population	$[0, 0.030]$	0.0031	0.0193
empirical, $N = 5000$	$[0, 0.034]$	0.0038	0.0210

The conditional covariance of the two accept indicators given a harmful label is 0.034 (0.037 empirically), and it is exactly 0 for the gluing filler. This is the circulation of [2, Prop. 12.3(iii)].

15.5 Horn data from separate studies

Take the parity 3-horn mixed with a restricted global law, with weight $t = 0.6$ on the parity part, so the population contextual fraction is 0.207. Each face is sampled in its own study of size N, over 50 replications (`horn_studies.csv`):

N per study	10^2	10^3	10^4	10^5
largest overlap disagreement (TV)	0.130	0.041	0.013	0.004
estimated contextual fraction	0.294	0.232	0.214	0.209

The incompatibility falls by $\sqrt{10}$ per decade, a primary obstruction caused by sampling. The contextual fraction converges to 0.207 from above, a secondary obstruction of the population.

A Changes from the earlier draft

Earlier claim	Status
22 chambers; 4/12/6 quadrilaterals, pentagons, hexagons	18 chambers; 2 triangles, 9 quadrilaterals, 6 pentagons, 1 hexagon (theorem 3.1). The accompanying script printed 18; a coordinate-order bug in its `intersect` produced wrong vertex counts.
Example 4.2 a quadrilateral	a triangle, $\mathbb{CP}^2$ (example 3.4)
Self-intersection table	corrected; the earlier patterns violated $\sum D_i^2 = 12 - 3k$
Flips; type change across $f_1 = \frac{1}{2}$	neither occurs (proposition 3.3)
Costs $c(\sigma, e_0) = c(e_0, \tau) = \frac{1}{2}$	both are 1; cost is a maximum over x
Postcomposition running h independently on equal pairs	not unital; replaced by definition 4.2. The earlier example then has zero defect, and new examples are nonzero (proposition 4.3)
Flattening surjective; Example 10.10 constant	not surjective; for point masses it is the identity (proposition 5.2)
Uniform level two monotone iff every cell is a facet; $n = 2$ not monotone	uniform $n \times n$ is degenerate; $n = 2$ is monotone, $n \geq 3$ is not simple
Level-two law restricted to couplings (Thm. 10.13)	ill-typed; replaced by proposition 5.2(v)
Facets are cycles of $K_{n,n}$; 15 facets; smooth Fano fourfold	facets are cells; 9 facets; uniform 3×3 is not simple; nondegenerate square $n \geq 3$ is not Fano (proposition 6.3)
Lattice distances are sums over cycles	they are the entries of the coupling at the fibre position
Critical points of W are vertices (committed states)	critical points are interior torus fibres with flat connections (proposition 6.1); vertices are fixed points and optimal-transport optima (proposition 6.4)
$F_2 = \operatorname{Hess} W$; Picard–Lefschetz formula; "symbolic proof"	withdrawn: F_2 is bilinear on couplings with values in circulations, $\operatorname{Hess} W$ is a quadratic form at a critical point, and the cited verification compared 0.1152 with 2. Replaced by $F_2 = \varepsilon b_2$ (section 8 and proposition 9.4)
Cycle counts $N_{2k} = \frac{1}{2k}\binom{n}{k}^2 k!(k-1)!$; $K_{4,4}$ total 156	$N_{2k} = \binom{n}{k}^2 k!(k-1)!/2$; $K_{4,4}$ has $36 + 96 + 72 = 204$; $K_{5,5}$ has $100 + 600 + 1800 + 1440$
Bridgeland 2008 for all surfaces; "(i)–(iii) none known"	Bridgeland 2008 is for K3; general surfaces are [1]; (i) is answered by proposition 7.1
Parity contextual fractions for $n = 3, 4$ only	$1 - 2^{2-n}$ for all n (section 13.1)

B Computational checks

The script `check_family.py` runs 5 checks (C1–C4), all passing, for proposition 12.1. The script `check_family_geometry.py` runs 6 more (G1–G3d): the hexagon vertex formulas, the exact area quadratics and the jump in the second derivative, and the Fisher determinant along the path.

`check_tower_fibration.py` runs 7 checks (W1–W7), all passing, for every entry of the table in section 11. The rows on support policies are checked by `check_support_floors.py` (SF1–SF8, all passing).

`check_nerve.py` runs 10 checks, all passing: A1–A2 for propositions 10.1 and 10.2, A3–A4 for proposition 10.3, A5–A6 for proposition 10.5 (A6 takes about a minute), and D1–D3 for proposi-

tion 9.4.

`check_cumulants.py` runs 8 checks for section 9, all passing: Q1–Q3 for propositions 9.1 and 9.2, and E1–E5 for proposition 9.3.

For sections 13 to 15, all passing:

- `check_horns_n.py` (4 checks, H1–H4) and `check_vgit.py` (4, V1–V3);
- `check_across_wall.py` (7, A1–A7) and `check_multiobject_cone.py` (11, M1–M11);
- `check_fukaya_probe.py` (6, K1–K6) and `fukaya_scan.py` (1, S1, by elimination at $\lambda = 500$ and 2000).
- For section 14: `check_tower_horns.py` (9, T1–T7).
- For section 15: `synthetic_examples.py` (8, Y1–Y8), which also writes the CSV files named in that section.

`check_polytope_companion.py` runs 26 checks, all passing:

- chambers, types, surfaces, the sum rule, orbits, the del Pezzo fan, wall-crossings and example 3.4 (F1–F7);
- propositions 2.2 and 3.5 in all 18 chambers (K1–K3);
- propositions 4.1 and 4.3 (S1–S4);
- propositions 5.1 and 5.2 (L1–L7);
- proposition 6.1, the non-monotone examples and proposition 6.3 (W1–W3);
- proposition 6.4 by linear programming (T1);
- the cycle counts of $K_{3,3}$ and $K_{4,4}$ by enumeration (C1).

Rational arithmetic is exact. The critical points are solved symbolically, and the linear programs use floating point.

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